PADHAKU
Exploration · Class 9 Science



CHAPTER 1

Exploration: Entering the World of Secondary Science

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Activity 1.1: Let us model

Activity 1.1 **Suppose you ride a bicycle from your school to your home. You want to model the time it takes to go home from school. What details would you keep? What details could you ignore? Suggest why ignoring some details may actually be useful.**

💡 SAMPLE ANSWER

Details to keep (they change the time a lot):

- Distance from school to home
- Your average speed on the bicycle
- Stops on the way (traffic signals, crossings, crowded roads)
- Uphill or downhill parts of the road

Details to ignore (very small or no effect):

- Colour or brand of the bicycle, colour of your school bag
- Small bends in the road, tiny changes in wind, exact tyre pressure

Why ignoring details is useful: the model becomes simple. We can quickly estimate time = distance ÷ average speed and still get a useful answer. Adding every small detail makes the model hard to use without making the answer much better.

Example: 3 km at about 12 km/h → 3 ÷ 12 = 0.25 h = **about 15 minutes** (plus a few minutes for stops).

⭐ Pause and Ponder

1 **Think of a prediction you or your family made recently (for example, the outcome of a cricket match). Was it based on evidence and reasoning, or mainly on guesswork? How can scientific thinking improve such predictions?**

💡 SAMPLE ANSWER

Example: We predicted India would win a cricket match because India is a "strong team". This was **mainly guesswork** (a feeling, not checked data).

Scientific thinking can improve it by:

- Using evidence: recent results of both teams, players' form, record at that ground
- Looking at conditions: pitch report, weather, who wins the toss
- Comparing with past patterns in similar matches
- Checking the prediction after the match and changing our reasons if it was wrong

⭐ Ready to Go Beyond · How much rice would feed a family of four for a month?

? **Assume all the family's calorie needs come from rice alone. An average adult needs about 2000–2500 kcal per day. Find out how many calories 100 g of uncooked rice provides when cooked and use this to estimate the rice needed by a family of four for a month.**

💡 SAMPLE ANSWER

100 g of uncooked rice gives about **350 kcal** (it becomes about 300 g of cooked rice).

Family need per day ≈ 4 × 2000 to 4 × 2500 = 8000 to 10,000 kcal (take about 9000 kcal).

Rice per day ≈ 9000 ÷ 350 × 100 g ≈ 2570 g ≈ 2.5 kg

Rice per month ≈ 2.5 kg × 30 ≈ **75 kg (roughly 70–85 kg)**

Check: this is much more than 100 g and much less than a few tonnes, so the estimate makes sense. (In real life families also eat dal, vegetables, milk etc., so they need less rice.)

⭐ Pause and Ponder

2 **Describe one situation where an approximate answer is good enough, and one where you would need a very exact value.**

💡 SAMPLE ANSWER

Approximate is enough: estimating how much rice to buy for a month, or how long it takes to walk to school. Being off by a little does not matter.

Exact value needed: a doctor giving the dose of a medicine, or the amount of fuel filled in an aeroplane. A small error can be dangerous.

3 **Choose a real-life object (maybe a pressure cooker or a mobile phone) or a problem (maybe a traffic jam near your school). Make a sketch listing what kind of ideas from physics, chemistry, biology, earth science, or mathematics are involved. Show how at least two branches of science connect with your example.**

💡 SAMPLE ANSWER

Example: Pressure cooker

- Physics: heat, steam pressure; higher pressure raises the boiling point of water, so food cooks faster; safety valve and whistle
- Chemistry: cooking changes the substances in food (starch softens, proteins change); steel body does not react with food
- Biology: cooked food is easier to digest; high temperature kills germs
- Earth science: in hills air pressure is low and water boils below 100 °C, so a pressure cooker is very useful there

- Mathematics: time and pressure settings, number of whistles, fuel saved

Connection: physics (higher pressure → higher boiling point) explains why the chemical changes of cooking (chemistry) happen faster, and this makes food soft and safe to eat (biology).

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**